

ARUNABH GHOSH

AI/ML Engineer

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✉ San Francisco Bay Area, California

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EXPERIENCE

Arista Networks

Data Scientist

📅 April 2021 – Present

📍 Santa Clara, USA

- Spearheaded an enterprise AI assistant built on a **multimodal agentic architecture** (patents pending) with **RAG** over 200,000+ technical documents and a custom **LLM-as-judge** evaluation pipeline, achieving **90%+ accuracy**.
- Engineered a **multi-model routing strategy** (GPT, Claude, Gemini) to balance cost and accuracy at scale; shipped a user-facing **React + FastAPI** interface now serving **15,000+ queries monthly** across **450+ weekly users**.
- Hardened the platform for enterprise deployment with **SSO**, **session management**, **NLP-based PII anonymization**, **CI/CD quality gates**, and **BigQuery analytics**; passed a comprehensive security and resilience audit.
- Authored **two granted U.S. patents**: **network fingerprint-based entity resolution** for endpoint tracking, and **device vulnerability risk scoring** fusing CVE, behavioral, and contextual data for security prioritization.

Chan Zuckerberg Biohub

Data Visualization Intern

📅 June 2020 – September 2020

📍 San Francisco Bay Area, USA

- Developed a **glyph-based algorithm** to convert anisotropy and orientation data from polarization microscopy into **intuitive tensor visualizations**.
- Improved processing speed by **500%** using **parallelization**, **vectorization**, & **precomputation**; used **PyVista** color schemes for enhanced interpretation.

EPFL

Research Intern

📅 June 2018 – July 2018

📍 Lausanne, Switzerland

- Implemented **Bayesian-optimized SGD** for ab-initio Cryo-EM protein reconstruction, outperforming established accuracy benchmarks.
- Applied **expectation-maximization** refinements, improving precision.

Google Summer of Code

Oppia Foundation, 7% acceptance rate

📅 May 2017 – August 2017

📍 Mumbai, India

- Engineered **full-stack learner dashboard** serving **400K+ users**, building high-throughput **MapReduce** pipeline for real-time tracking with personalized progress visualization, feedback management, and bookmarks.

PROJECTS

Pathways: Branching AI Conversations

Open Source, 2025

- Built a **React/Express** app for non-linear AI conversations using a **graph-based message structure** for instant, zero-duplication branching; applied a **dual-model strategy** (Sonnet + Haiku) to balance quality and cost.

Multi-Agent Financial Planning

Open Source, 2025

- Designed a **4-agent orchestration pipeline** (search, code, execution, advisory) using **MagenticOne's dual-ledger planning**; deployed a **multi-model architecture** for financial research, projections, and portfolio optimization.

Molecular Structure Estimation, Thesis

Prof. Ajit Rajwade, IIT-B

- Developed a pipeline integrating **hierarchical clustering**, **graph-Laplacian** embeddings, and sparse optimization to reconstruct protein structures from noisy, outlier-prone data, with no prior info required. **ICIP, 2019**

Efficient Training via Subset Selection

Prof. B. Mirzasoleiman, UCLA

- Implemented **gradient-based selection methods** to reduce CIFAR-10 training set by **40%** while maintaining accuracy through augmentation.

EDUCATION

UCLA

Master of Science

📅 2019 – 2021

📍 USA

- Electrical and Computer Engineering | GPA: **3.97**

IIT Bombay

Bachelor of Technology

📅 2015 – 2019

📍 India

- Electrical Engineering (Hons) | GPA: **8.84/10**
- Minor Degree in Computer Science

PUBLICATIONS & PATENTS

- Ghosh, A., "Network endpoint identification through network fingerprint based entity resolution," **U.S. Patent No. 11,799,882**, Oct. 24, 2023. Assignee: Arista Networks Inc.
- Ghosh, A., "Device vulnerability risk assessment system," **U.S. Patent No. 12,598,209**, Apr. 7, 2026. Assignee: Arista Networks Inc.
- Ghosh, A., et al., "Ab initio tomography with object heterogeneity and unknown viewing parameters," in **IEEE ICIP, 2019. (Oral)**

SKILLS

🔧 Programming & Tools

- **Languages:** Python (Primary), SQL, Scala
- **GenAI:** LLMs, RAG, Agentic AI, LangChain, LlamaIndex, Hugging Face, FAISS
- **ML & Data:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BigQuery
- **Web & Cloud:** React, FastAPI, Streamlit, GCP (Vertex AI, Firestore), Azure, Docker

🎓 Academic Courses

- **ML & AI:** Neural Networks & Deep Learning (A+), Reinforcement Learning
- **Data Science:** Large-Scale Data Mining, Matrix Analysis (A+), Big Data Systems
- **Image & Speech:** NLP, Image Processing & Computer Vision, Computational Imaging

📖 Teaching Experience (UCLA)

- Mathematics for Life Scientists (UG)
- Feedback for Control Systems (Grad)
- Signals and Systems (Grad)
- Neural Networks & Deep Learning (Grad)

ACHIEVEMENTS

- All India Rank of **135** in JEE Advanced 2015
- **Silver Medal** at the 6th Inter IIT Technical Meet
- **Best Tech Prize**, Yahoo Hackathon, AR Game